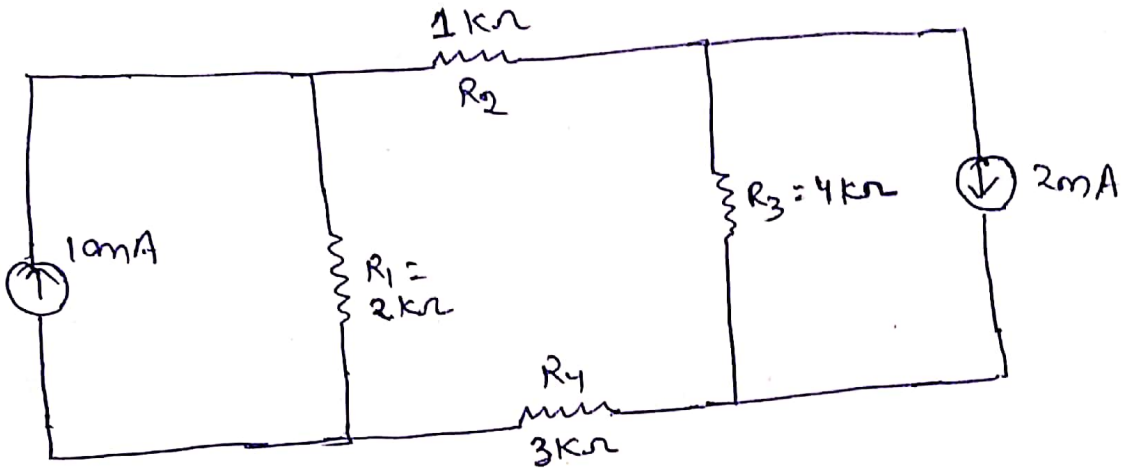
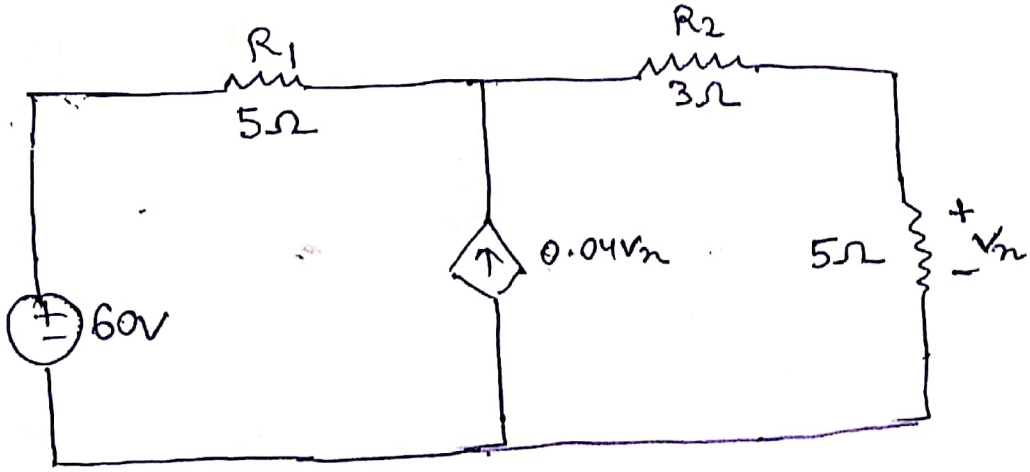
ET Assignment 1

Q1 Using Mesh Analysis find the magnitude of current (in mA) through the resistor R_2 in the figure shown



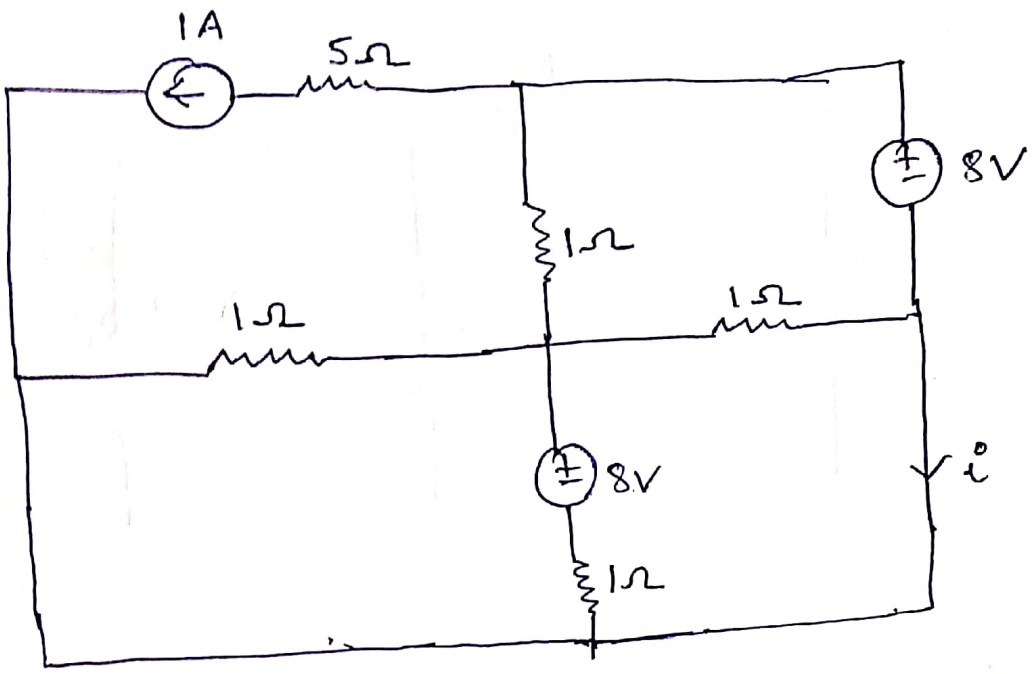
Q2. For the figure shown in the figure, find the magnitude of the current (in amperes) through R_2. Use Mesh Analysis.



Scanned by CamScanner

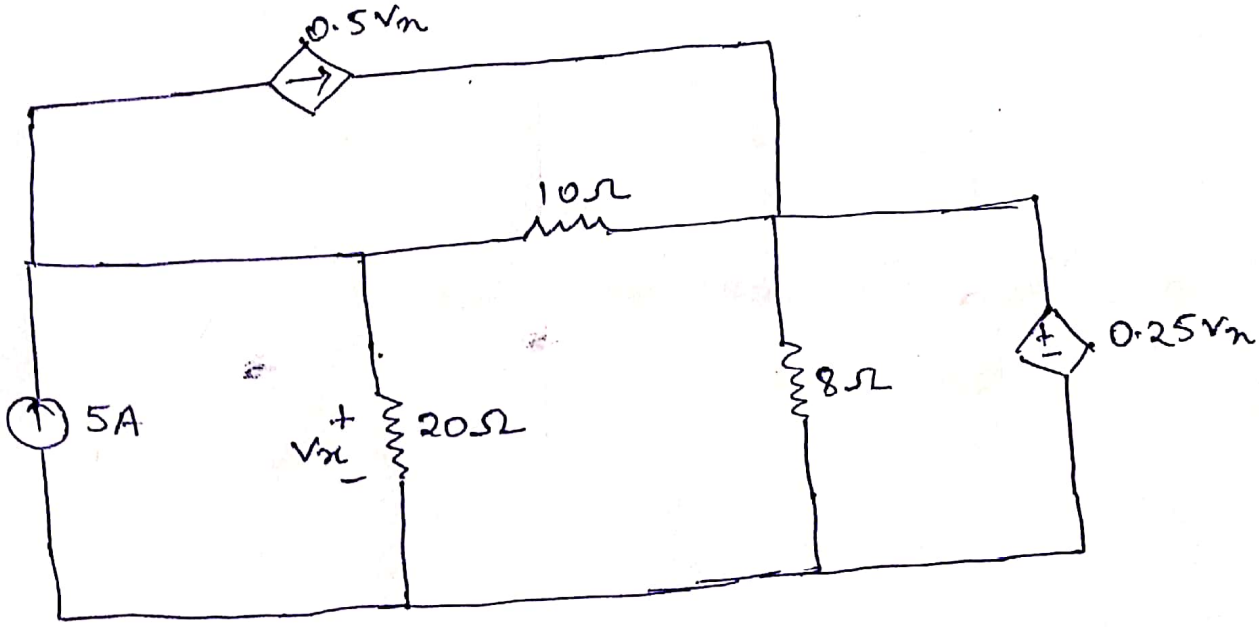
Q3 For the given figure, Find the current i.

Use nodal analysis



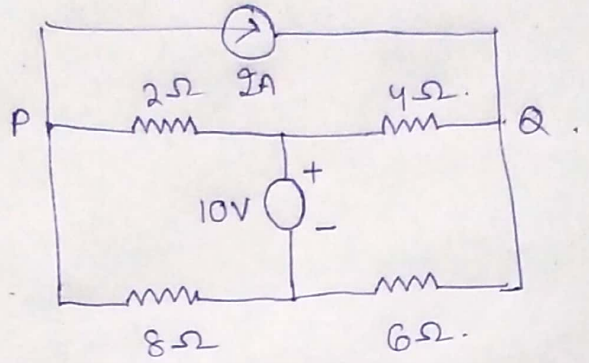
Q4. In the circuit shown, Find the voltage 'V_x' in volts.

Use nodal analysis

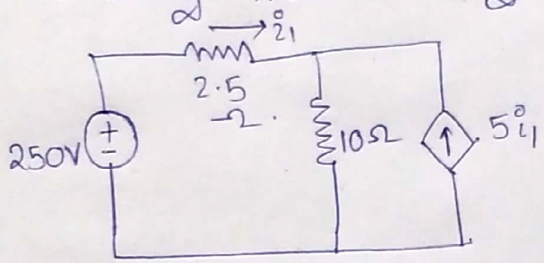


Scanned by CamScanner

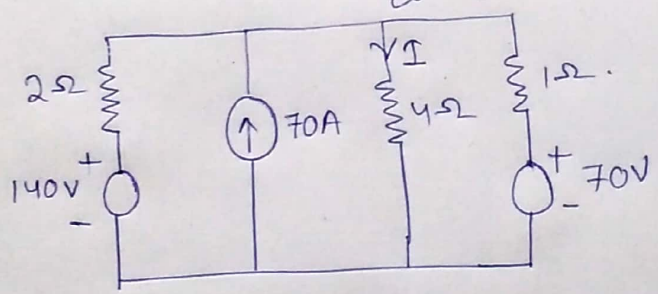
Q5 In the given figure, the potential difference betn points P & Q is —.
Solve it using mesh analysis.



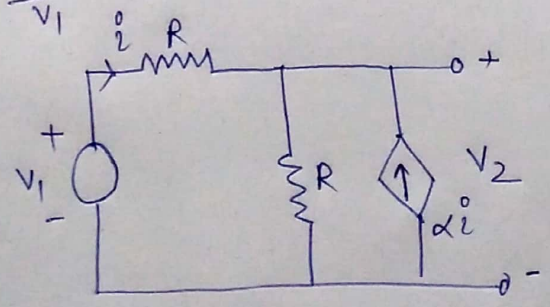
Q6 Calculate the value of i_1 in the shown figure. (Using mesh analysis)



Q7 What is the value of current I in the circuit shown? (Using nodal analysis)

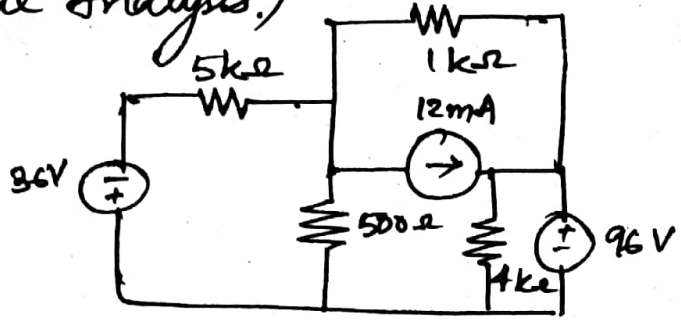


Q8 Consider the circuit as shown in figure, which has a current dependent current source. The value $\frac{V_2}{V_1}$ is

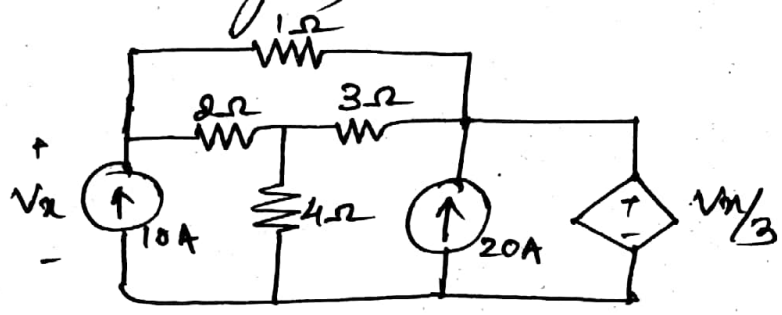


Scanned by CamScanner

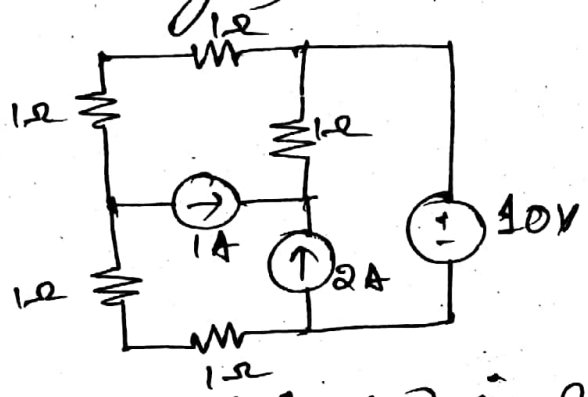
Q9 Find the voltage across 500Ω resistor.
(Use Nodal Analysis)



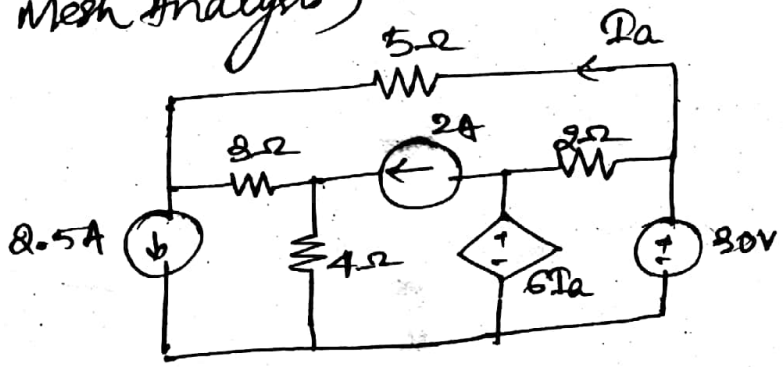
Q10 In the ckt below find the value of V_x, currents flowing through 1Ω resistor.
(Use Nodal Analysis)



Q11 In the ckt below, find the power delivered/absorbed by the battery.
(Use Mesh Analysis)

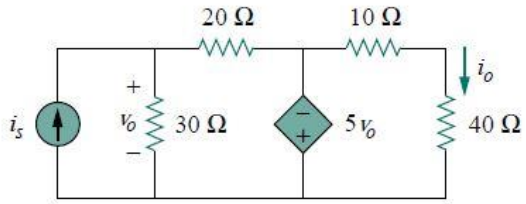


Q12 Find the power dissipated in 2Ω resistor
(use Mesh Analysis)

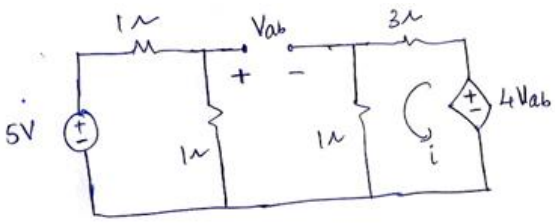


Scanned by CamScanner

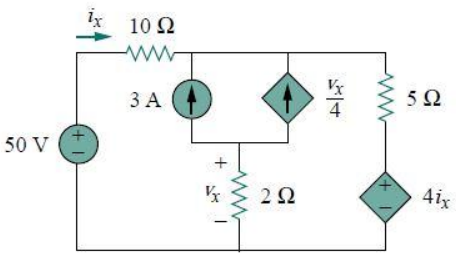
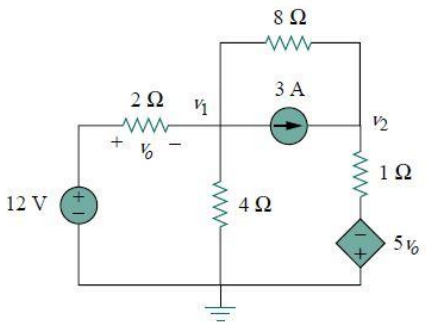
13. Using mesh analysis determine the current gain i_o / i_s in the given circuit.



14. In the circuit given below determine the current *i*



15. Find v_x and i_x in the given circuit using mesh analysis



16. Determine the voltages v1 and v2 shown in the figure using nodal analysis